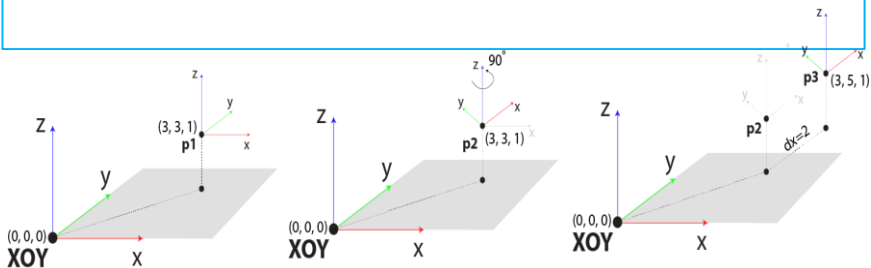


ParaPy Positioning Cheatsheet

Point(x=0, y=0, z=0)

```
>>> pt1 = Point(1, 2, 3)
>>> pt2 = Point(3, 4, 5)
>>> pt1[0]
1
>>> pt1.x
1
>>> for i in pt1:
    print i,
1 2 3
>>> list(pt1)
[1, 2, 3]
>>> -pt1
Point(-1, -2, -3)
>>> pt2 - pt1
Vector(2, 2, 2)
>>> pt1 - pt2
Vector(-2, -2, -2)
>>> pt1.distance(pt2)
3.46...
>>> pt1.midpoint(pt2)
Point(2, 3, 4)
>>> pt1.interpolate(pt2, 0.75)
Point(2.5, 3.5, 4.5)
>>> pt1.translate(x=1)
Point(2, 2, 3)
>>> pt1.rotate('z', pi/2)
>>> pt1.rotate('z', 90, deg=True)
>>> pt1.rotate90('z')
Point(-2, 1, 3)
>>> pt1.rotate_around(pt2, 'z', pi/2)
>>> pt1.rotate_around(pt2, 'z', 90, deg=True)
>>> pt1.rotate90_around(pt2, 'z')
Point(5, 2, 3)
>>> pt1.project(pt2, 'z')
Point(3, 4, 3)
>>> pt1.project(pt2, 'y', 'z')
Point(3, 2, 3)
>>> pt1.polygon('x', 1, 'y', 2)
[Point(1, 2, 3), Point(2, 2, 3), Point(2, 4, 3)]
>>> ORIGIN
Point(0, 0, 0)
```



```
>>> p1 = translate(XOY, x=3, y=3, z=1)
>>> p1.location
Point(3, 3, 1)
>>> p1.Vx
Vector(1, 0, 0)
>>> p2 = rotate90(p1, 'z')
>>> p2.Vx
Vector(0, 0, 1.0, 0.0)
>>> p3 = translate(p2, x=2)
>>> p3.location
Point(3.0, 5.0, 1.0)
>>> p3.Vx
Vector(0, 0, 1.0, 0.0)
```

Position(location=ORIGIN, orientation=XY)

```
>>> p1 = Position(pt1, XY)
>>> p2 = Position(pt1, YZ)
>>> p1[0], p1.x
1, 1
>>> p1.location
Point(1, 2, 3)
>>> p2.orientation
Orientation(x=Vector(0.0, 1.0, 0.0),
           y=Vector(0.0, 0.0, 1.0),
           z=Vector(1.0, 0.0, 0.0))
>>> p1.Vx, p1.Vy
(Vector(1, 0, 0), Vector(0, 1, 0))
>>> p2.Vx, p2.Vy
(Vector(0, 1, 0), Vector(0, 0, 1))
>>> p1.translate(x=1), p2.translate(x=1)
(Position(2, 2, 3), Position(1, 3, 3))
>>> p1.get_point(x=1), p2.get_point(x=1)
(Point(2, 2, 3), Point(1.0, 3.0, 3.0))
>>> XOY, YOZ, ZOX
...
```

translate(p, v1, s1[, v2, s2, ...])

```
>>> pt = Point(1, 0, 0)
>>> translate(pt, 'x', 1)
Point(2, 0, 0)
>>> translate(pt, x=1)
Point(2, 0, 0)
>>> translate(pt, Vector(1, 0, 0), 1)
Point(2, 0, 0)
>>> translate(pt, 'x', 1, 'y', 1)
Point(2, 1, 0)
>>> pos = Position(pt).rotate90('z')
>>> translate(pos, 'x', 1)
Position(1, 1, 0)
>>> translate(pos, x=1)
Position(1, 1, 0)
>>> translate(pos, Vector(1, 0, 0), 1)
Position(2, 0, 0)
>>> translate(pos, 'x', 1, 'y', 1)
Point(0, 1, 0)
```

rotate(p, dir, angle[, ref=q, deg=bool])
rotate90(p, dir, [, ref=q, deg=bool])

```
>>> pt = Point(1, 0, 0)
>>> rotate(pt, 'z', pi/2.)
Point(0, 1, 0)
>>> rotate(pt, 'z', radians(90))
Point(0, 1, 0)
>>> rotate(pt, 'z', 90, deg=True)
Point(0, 1, 0)
>>> pt = Point(1, 0, 0)
>>> rotate90(pt, 'z')
Point(0, 1, 0)
>>> rotate90(pt, 'z', ref=Point(0.5, 0, 0))
Point(0.5, 0.5, 0.0)
```